10-152-118 HTML, CSS & JavaScript

Course Outcome Summary

COURSE INFORMATION

Description:

Learners develop foundational web development skills using HTML, CSS, and JavaScript to create structured, styled, and interactive web pages. The course emphasizes building multi-page websites, applying design principles for usability and readability, and implementing basic interactivity through JavaScript. Learners practice debugging and refining web applications using progressive techniques and are introduced to AI-assisted tools to support development. By the end of the course, students are able to design, build, and explain functional web solutions.

Career Cluster: Information Technology

Instructional Level: A.A.S. - Associate in Applied Science

Total Credits: 3

Total Hours: 72

TEXTBOOKS

Responsive Web Design with HTML5 and CSS: Build future-proof responsive websites using the latest HTML5 and CSS techniques - 5th ed. Edition - Ben

JavaScript and jQuery: Interactive Front-End Web Development - 1st Edition - Jon Duckett

HTML 5: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

CSS - Cascading Style Sheets: a QuickStudy Laminated Reference Guide *Pamphlet* – Robin Nixon (Optional)

COURSE COMPETENCIES

1. Structure web content using HTML

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

1.1. Project assignments and quiz

Criteria

1.1. Create structured web pages using semantic HTML elements

1.2. Organize content logically using headings, paragraphs, lists, and links

1.3. Build multi-page website structures

1.4. Apply consistent page organization

1.5. Ensure structure supports styling and interactivity

Learning Objectives

1.a. Define HTML structure and purpose

1.b. Identify common HTML elements
1.c. Explain semantic markup
1.d. Describe how HTML structure supports styling and interaction

2. Design and style web pages using CSS to create readable, usable, and responsive layouts

Domain: Cognitive Level: Creating Status: WIP

Assessment Strategies

2.1. Project assignments and quiz

Criteria

2.1. Apply CSS to control layout, spacing, typography, and color
2.2. Use selectors, classes, and IDs effectively
2.3. Create visually consistent and readable designs
2.4. Apply basic responsive design techniques
2.5. Ensure styling supports usability and interaction

Learning Objectives

2.a. Define CSS styling principles
2.b. Explain layout concepts (box model, positioning)
2.c. Identify responsive design basics
2.d. Explain how styling impacts usability and user experience

3. Implement interactive behavior using JavaScript to respond to user input and manipulate web page content

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

3.1. Project assignments and quiz

Criteria

3.1. Write basic JavaScript using variables, conditionals, and functions
3.2. Implement interactive behaviors in web pages
3.3. Respond to user input and events
3.4. Connect program logic to visible outcomes
3.5. Manipulate HTML elements and styles dynamically

Learning Objectives

3.a. Define JavaScript as a programming language
3.b. Explain variables, functions, and control flow
3.c. Describe how JavaScript enables interactivity
3.d. Explain how JavaScript interacts with HTML and CSS

4. Debug and evaluate web applications using progressive techniques across HTML, CSS, and JavaScript

Domain: Cognitive Level: Analyzing Status: WIP

Assessment Strategies

4.1. Project assignments and quiz

Criteria

4.1. Identify and correct layout and styling issues

4.2. Use console logging to trace JavaScript behavior
4.3. Utilize browser tools to inspect and troubleshoot web pages
4.4. Apply structured debugging approaches to resolve errors
4.5. Diagnose issues across HTML, CSS, and JavaScript interactions

Learning Objectives

4.a. Define debugging and troubleshooting
4.b. Identify common HTML, CSS, and JavaScript issues
4.c. Explain debugging strategies
4.d. Describe how issues can originate across multiple layers

5. Develop web solutions using structured problem-solving, appropriate tools, and iterative improvement

Domain: Cognitive Level: Creating Status: WIP

Assessment Strategies

5.1. Project assignments and quiz

Criteria

5.1. Plan and build web pages based on defined requirements
5.2. Apply structured problem-solving to design and implementation
5.3. Use AI tools to assist in development when appropriate
5.4. Evaluate and refine solutions for usability and clarity
5.5. Iteratively improve solutions based on testing and feedback

Learning Objectives

5.a. Describe structured development approaches
5.b. Identify when and how to use AI tools
5.c. Explain iterative improvement of web solutions
5.d. Explain how planning, building, and refining form a development cycle